

A Byte-Level Multilingual News Recommender for Microcontrollers

Abstract. Running a news recommender on the device itself keeps a reader's clicks private, but the usual models store a word-embedding table of tens to hundreds of megabytes per language, which no microcontroller's flash can hold. We avoid the table by reading a title as its raw UTF-8 bytes: a small char-CNN whose 256-value vocabulary is shared by every language. Around this encoder we search three families of architecture (unconstrained, budget-constrained and binary-aware) and evaluate each at full precision and with inner-layer weights at eight bits and at one bit, measuring ranking quality on MIND against estimated flash, RAM and energy. With eight-bit weights the constrained model scores 0.610 AUC at an estimated 204 KB and the unconstrained one 0.647 at 790 KB. For reference, a reduced NRMS with a 27 MB word table scores 0.607 and a per-user category histogram with no network 0.598. Initialising the encoder by distillation from a frozen multilingual sentence-transformer adds 3.3 AUC points. A naive one-bit model stays near chance (0.521); ReActNet blocks started from the distilled weights bring it to 0.572 at 194 KB. Seeded reruns differ from the matrix run by up to 0.02 AUC, so smaller gaps are not resolved. The matrix models were trained on English clicks only and transfer little to the fourteen xMIND translations (0.522 on average). **[TODO: per-language results of the distilled encoder; latency and energy on STM32 boards and an Android phone].** Code and artefacts are released.

Keywords: Edge AI · Neural Architecture Search · Binarized Neural Networks · Multilingual News Recommendation · TinyML · Knowledge Distillation

1 Introduction

Running personalisation on the phone or sensor node rather than the cloud is attractive for privacy: the record of what a user reads never leaves the device. Two difficulties come with that setting. A model that starts from nothing for each new user recommends poorly until it has seen enough clicks, and the standard neural recommenders (NRMS [21], NAML [20] and their relatives) represent a title with a word-embedding matrix. That matrix is large: GloVe-300d over 30–100k words is already 37–120 MB, one table per language, and none of it fits a microcontroller's few megabytes of flash.

We study both problems offline, on MIND [23] and its translated extension xMIND [6], and then shrink the result for the device. The idea is to drop words

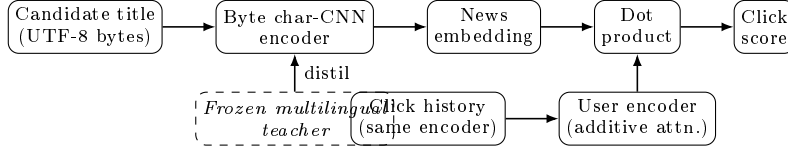


Fig. 1. The recommender: a byte char-CNN embeds candidate and history titles; an additive-attention encoder pools the history and a dot product scores the candidate. A frozen multilingual teacher provides distillation targets. Only the char-CNN is searched and quantised.

and read a title as its UTF-8 bytes. A byte takes 256 values, so the vocabulary is a single 256-row table shared by every language, and one small convolutional encoder reads Chinese, Georgian and English alike. A frozen multilingual teacher [17] supplies distillation targets that place a translated title near its English original; we use them to score candidate architectures and to initialise the encoder.

We contribute three things. A byte-level content encoder removes the embedding-table memory wall: its vocabulary is one 256-row table for every language. Around it we benchmark three architecture searches (unconstrained, budget-constrained and binary-aware) at three weight precisions, reporting accuracy alongside estimated flash, RAM and energy. Finally we bring the one-bit case from near chance to 0.572 AUC with ReActNet activations and Bi-Real shortcuts started from distilled weights. Character-level inputs have been used for click prediction [3], and vocabulary-free projections give text classifiers of about 200KB [7]. A search of arXiv and IEEE Xplore (October 2026) returned no news recommender that reads raw bytes and none evaluated on a microcontroller; the nearest on-device NAS and binary work is on vision [9, 10, 15].

2 Related Work

Neural news recommendation on MIND is well charted. NRMS [21], NAML [20], LSTUR, NPA and DKN encode titles and users with attention or knowledge graphs; their published test AUC on the large split runs from 0.65 (DKN) to 0.68 (NRMS) [23]. Language-model recommenders (PLM-NR, Fastformer, UNBERT) reach 0.70–0.73 [22] but wrap a BERT-sized encoder of hundreds of megabytes, out of reach on a microcontroller; we keep NRMS as the reference that is closest in size. The tools we borrow are vocabulary-free character and byte models [24, 4, 2], cross-lingual distillation [17], memory-budgeted NAS [9], and the binary-network line of XNOR-Net [16], Bi-Real [12] and ReActNet [11]; what is new is putting them together for on-device multilingual news.

3 Method

We treat recommendation as impression-level ranking: score candidate titles against the titles a user has clicked, embedding both with the same encoder (Fig. 1). The encoder reads a title’s bytes through a 256-entry embedding, a few depthwise-separable one-dimensional convolutions, a masked average over positions, and a linear projection. An additive-attention encoder summarises the user’s history, and the score is the dot product of the two vectors, trained with one positive and four negatives per step (AdamW). A frozen `paraphrase-multilingual-MiniLM-L12-v2` [19] supplies a 384-dimensional target for each English title, and because the xMIND titles are translations of those same articles the distillation loss maps every translation to that one target; a Matryoshka [8] term keeps the leading coordinates informative on their own. Distillation has two uses here: it is the fitness of the architecture search, and it initialises the encoder in the ablation and in the one-bit model of Sect. 5. The nine models of Table 1 are trained on clicks from a random initialisation.

The search has three arms: one unconstrained, one held to an on-device budget with eight-bit weights, and one held to the same budget with one-bit weights. The space holds 120 architectures (six channel counts, five depths, four output widths). Each arm runs an evolutionary search over it, scores a candidate by a short distillation, and rejects candidates over budget; precision is then the second axis. Quantisation is simulated on the weights with a straight-through estimator, per-channel for eight bits and sign-times-scale for one. Following XNOR-Net, the byte embedding and the outermost layers stay in full precision, and so do all activations, so the quantised rows cover 66%, 31% and 24% of the parameters of the three architectures. For the binary arm alone we add the ReActNet activations and Bi-Real shortcuts, binarise the block inputs and start from the distilled weights, which is what makes one-bit training converge to something useful. Sizes are parameter counts at each layer’s precision and multiply-accumulates are counted per layer. The energy column multiplies them by per-operation figures: 4.6 and 0.23 pJ for a 32-bit float and an eight-bit integer multiply-accumulate at 45 nm [5], and one sixty-fourth of the former for a one-bit operation, the operation-count convention of binary networks [16, 12]. We read it only as a relative ordering, since it ignores the memory traffic that often dominates on a real chip.

4 Experimental Setup

We use MINDsmall [23]: fifty thousand users, about 157k training and 73k development impressions over 51k articles, together with the fourteen NLLB translations of xMIND [6, 14]. xMIND provides only translated titles, keyed to the same article ids, so we reuse the English behaviour logs and swap the text. Every recommender we train, and the NRMS baseline, sees only English clicks, and the other languages appear only at evaluation. The multilingual numbers are therefore a test of cross-lingual transfer in the content encoder, not of native-language

Table 1. Architecture $\times$ weight precision on MINDsmall dev, single runs. **Bold:** best per group; †: precision the arm was searched at. Arm = the search, precision = simulated weight quantisation, so off-diagonal cells are ablations. Size and energy are estimates (Sect. 3).

Arm	Prec.	AUC	MRR	nDCG@10	Size (KB)	Energy (μ J)
NAS	FP32 [†]	0.633	0.326	0.375	1566.8	166.3
NAS	INT8	0.647	0.338	0.387	789.8	17.91
NAS	Binary	0.588	0.297	0.342	563.1	12.54
Micro-NAS	FP32	0.605	0.307	0.352	266.8	15.15
Micro-NAS	INT8 [†]	0.610	0.314	0.360	203.9	3.16
Micro-NAS	Binary	0.521	0.251	0.294	185.6	2.72
Bin. μ NAS	FP32	0.593	0.290	0.336	311.4	14.98
Bin. μ NAS	INT8	0.601	0.295	0.344	255.7	4.35
Bin. μ NAS	Binary [†]	0.546	0.273	0.317	239.5	3.96

personalisation. We stay on the small split rather than the million-user MIND-large because the full study — three arms, three precisions, the search and the fifteen-language evaluation — already takes 8.2 hours on one laptop; the small split has no public test set, so we evaluate on dev with the number of epochs fixed in advance. The search runs a population of thirty over fifteen generations, scoring each candidate by a two-epoch distillation on twenty thousand titles. The budgets of the constrained arms are 256 KB of flash and 128 KB of RAM, about eight times tighter than an STM32H7 holds. A bound of 2 M multiply-accumulates was also set, but the counter of that run skipped the quantised layers, so both constrained winners (3.29 M and 3.26 M) exceed it; **[TODO: re-run the constrained search with the corrected count]**. Everything runs on one laptop: an Intel Core Ultra 9 275HX with an eight-gigabyte RTX 5070 Laptop GPU and 32 GB of memory, PyTorch 2.12 on CUDA 13. The intended targets are STM32 microcontrollers and an Android phone; **[TODO: board and phone measurements]**, and this draft reports laptop timings and analytical estimates only. Negative sampling and the search are seeded; network initialisation in the matrix run was not, which the seeded reruns of Sect. 5 quantify. The downloaded data is pinned by checksum, and the code, notebook, result files and a dependency lock are released.¹

5 Results

Table 1 collects the nine combinations (the three arms found the architectures 256-4-384, 64-5-384 and 96-2-384, with 36.1, 3.29 and 3.26 million multiply-accumulates per title). Two patterns emerge. Moving the inner weights from full to eight-bit precision changes AUC by at most 0.014, inside the run-to-run variation reported below, while the energy estimate falls 3.4 to 9.3 times and the size 1.2 to 2 times. One bit is different: it loses five to nine AUC points, and because the embedding and outer layers stay in full precision its size advantage over eight bits is small at this scale. The RAM estimate, the input and output

¹ <https://github.com/SepehrMohammady/MIND-Edge-Recommender>

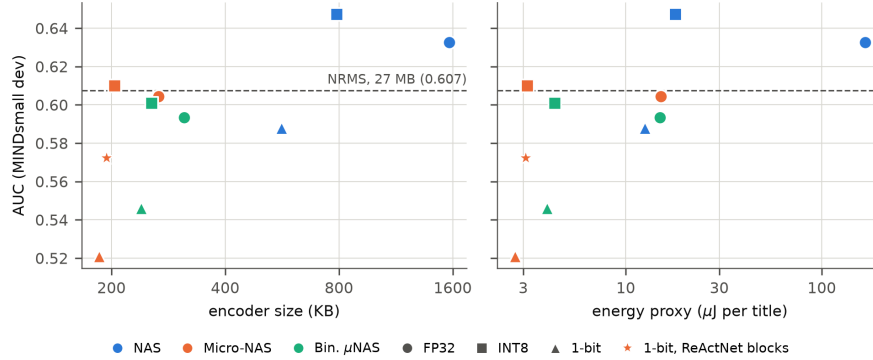


Fig. 2. Accuracy against estimated encoder size (left) and against the energy estimate (right), per arm and weight precision; dashed line: reduced NRMS with a 27 MB word table.

buffer of the widest layer at one byte per activation, is 16 KB for the constrained model and 64 KB for the unconstrained one. On the laptop a title takes 0.10–0.34 ms on CPU (onnxruntime) and under 0.7 ms on GPU for the three full-precision architectures.

Three reference points place these numbers. Our NRMS reproduction, a reduced variant with a randomly initialised 256-dimensional word table, reaches 0.607; 92% of its 7.08 million parameters are the table (27 MB, or 6.8 MB in eight bits at the same accuracy). NRMS with GloVe vectors is reported at 0.656 on MIND-small [13]; **[TODO: our GloVe rerun]**. Two scorers with no network set a floor: popularity gives 0.525, and a histogram of the reader’s clicks over MIND’s category labels 0.598, or 0.617 with the finer subcategory labels. The eight-bit constrained encoder (0.610) is thus level with the reduced NRMS at about 135 times less storage, but also within 0.02 of a label histogram; the unconstrained one (0.647) clears both. Feeds outside MIND carry no such editorial labels, which is the case a content encoder is for.

The matrix models saw only English bytes. Read through the full-precision constrained model, the fourteen translated dev sets score between 0.491 (Georgian) and 0.556 (Haitian Creole), 0.522 on average, against 0.605 in English: without multilingual training almost nothing transfers. The distilled encoder is the one built for this; **[TODO: its per-language scores]**.

Three further measurements qualify the table. Starting the encoder from distilled weights rather than from scratch raises AUC from 0.600 to 0.633 at a fixed architecture (64-5-384, full precision), so the gain comes from distillation, not capacity. Three seeded reruns of the eight-bit constrained model score 0.630, 0.622 and 0.621; the matrix run, whose initialisation was not seeded, scored 0.610, so we read gaps below about 0.02 in Table 1 as unresolved. For the 3% of impressions whose reader has no history, a history-based score ties every candidate (0.500 by construction); popularity gives 0.529 there and the population’s category distribution 0.533.

Binarising the inner weights the naive way leaves the constrained model near chance (0.521). Replacing its blocks with full binary convolutions, ReActNet activations and Bi-Real shortcuts, and starting from distilled weights, gives 0.572 at an estimated 194KB, 0.035 below the reduced NRMS. The 0.05 gain exceeds the run-to-run variation; three seeded runs of this model on a reduced training budget have a standard deviation of 0.012. On the accuracy-versus-footprint view (Fig. 2) eight-bit Micro-NAS (0.610, 204KB, 3.2 μ J) has the lowest cost at the accuracy of the reduced NRMS, while unconstrained eight-bit leads on accuracy (0.647, 790KB).

6 Limitations

The main gap is hardware. We timed the full-precision models on the laptop and estimated size, RAM and energy from parameter and operation counts, but ran nothing on a microcontroller or a phone. The quantisation is simulated on weights only: activations, the byte embedding and the outer layers stay in floating point, so the eight-bit and one-bit rows may overstate accuracy and their sizes are not those of an integer-only file; **[TODO: full-integer export, with accuracy measured from the exported model]**. Eight-bit inference has a direct path to STM32 boards through quantised ONNX or TFLite models and ST Edge AI. The one-bit path needs a binary runtime such as CBin-NN [18] or Larq Compute Engine [1], and the Cortex-M7 has no popcount instruction, so we make no speed claim for binary and treat it only as a way to save space. The energy figure ignores SRAM and memory traffic, which can dominate on a Cortex-M. The constrained search enforced its flash and RAM budgets but not the operation budget, the NRMS reference is a reduced variant, and the cross-lingual result so far is negative. Finally, a byte sequence is about five times longer than the word sequence, trading flash for computation, and all of this is on the small MIND split.

7 Ethics and Conclusion

MIND and xMIND are used under their non-commercial research licences; the work touches no personal data beyond the public logs, and the translations are machine-generated. Replacing the word-embedding table with a byte-level encoder removes the largest item of a news recommender’s storage and gives every language the same 256-row vocabulary. With eight-bit weights the constrained encoder reaches 0.610 AUC at an estimated 204KB; the one-bit model, started from distilled weights, reaches 0.572 at 194KB. What is still missing: **[TODO: per-language results of the distilled encoder, a full-integer model, board and phone measurements, and a one-bit runtime]** — that is where we take it next.

References

1. Bannink, T., Bakhtiari, A., Hillier, A., et al.: Larq compute engine: Design, benchmark, and deploy state-of-the-art binarized neural networks. In: Proceedings of MLSys (2021), arXiv:2011.09398
2. Clark, J.H., Garrette, D., Turc, I., Wieting, J.: CANINE: Pre-training an efficient tokenization-free encoder for language representation. In: Transactions of the ACL (2022)
3. Edizel, B., Mantrach, A., Bai, X.: Deep character-level click-through rate prediction for sponsored search. In: Proceedings of SIGIR. pp. 305–314 (2017). <https://doi.org/10.1145/3077136.3080811>
4. Gillick, D., Brunk, C., Vinyals, O., Subramanya, A.: Multilingual language processing from bytes. arXiv:1512.00103 (2016)
5. Horowitz, M.: Computing’s energy problem (and what we can do about it). In: IEEE ISSCC. pp. 10–14 (2014)
6. Iana, A., Glavaš, G., Paulheim, H.: MIND your language: A multilingual dataset for cross-lingual news recommendation. In: Proceedings of SIGIR (2024). <https://doi.org/10.1145/3626772.3657867>
7. Kaliamoorthi, P., Ravi, S., Kozareva, Z.: PRADO: Projection attention networks for document classification on-device. In: Proceedings of EMNLP-IJCNLP (2019). <https://doi.org/10.18653/v1/D19-1506>
8. Kusupati, A., Bhatt, G., Rege, A., Wallingford, M., Sinha, A., Ramanujan, V., Howard-Snyder, W., Chen, K., Kakade, S., Jain, P., Farhadi, A.: Matryoshka representation learning. In: Proceedings of NeurIPS (2022)
9. Lin, J., Chen, W.M., Lin, Y., Cohn, J., Gan, C., Han, S.: MCUNet: Tiny deep learning on iot devices. In: Proceedings of NeurIPS (2020)
10. Lin, Z., Wang, Y., Zhang, J., Chu, X., Ling, H.: NAS-BNN: Neural architecture search for binary neural networks. Pattern Recognition **159**, 111086 (2025). <https://doi.org/10.1016/j.patcog.2024.111086>
11. Liu, Z., Shen, Z., Savvides, M., Cheng, K.T.: ReActNet: Towards precise binary neural network with generalized activation functions. In: Proceedings of ECCV (2020)
12. Liu, Z., Wu, B., Luo, W., Yang, X., Liu, W., Cheng, K.T.: Bi-real net: Enhancing the performance of 1-bit cnns with improved representational capability and advanced training algorithm. In: Proceedings of ECCV (2018)
13. Mao, Z., Li, J., Wang, H., Zeng, X., Wong, K.F.: DIGAT: Modeling news recommendation with dual-graph interaction. In: Findings of EMNLP (2022), arXiv:2210.05196
14. NLLB Team: No language left behind: Scaling human-centered machine translation (2022)
15. Pighetti, A., Bellotti, F., Mohammady, S., Lazzaroni, L., Ballout, H., Berta, R.: Exploring neural architecture search (NAS) for binary neural networks (BNNs) in the new WakeVision. In: Applications in Electronics Pervading Industry, Environment and Society (ApplePies), Lecture Notes in Electrical Engineering, vol. 1553, pp. 85–90 (2026). https://doi.org/10.1007/978-3-032-17174-0_12
16. Rastegari, M., Ordonez, V., Redmon, J., Farhadi, A.: XNOR-Net: Imagenet classification using binary convolutional neural networks. In: Proceedings of ECCV (2016)
17. Reimers, N., Gurevych, I.: Making monolingual sentence embeddings multilingual using knowledge distillation. In: Proceedings of EMNLP (2020)

18. Sakr, F., Berta, R., Doyle, J., Capello, A., Dabbous, A., Lazzaroni, L., Bellotti, F.: CBin-NN: An inference engine for binarized neural networks. *Electronics* **13**(9), 1624 (2024). <https://doi.org/10.3390/electronics13091624>
19. Wang, W., Wei, F., Dong, L., Bao, H., Yang, N., Zhou, M.: MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers. In: *Proceedings of NeurIPS* (2020)
20. Wu, C., Wu, F., An, M., Huang, J., Huang, Y., Xie, X.: Neural news recommendation with attentive multi-view learning. In: *Proceedings of IJCAI* (2019)
21. Wu, C., Wu, F., Ge, S., Qi, T., Huang, Y., Xie, X.: Neural news recommendation with multi-head self-attention. In: *Proceedings of EMNLP-IJCNLP*. pp. 6389–6394 (2019)
22. Wu, C., Wu, F., Huang, Y., Xie, X.: Personalized news recommendation: Methods and challenges. *ACM Transactions on Information Systems* (2023), arXiv:2106.08934
23. Wu, F., et al.: MIND: A large-scale dataset for news recommendation. In: *Proceedings of ACL*. pp. 3597–3606 (2020)
24. Zhang, X., Zhao, J., LeCun, Y.: Character-level convolutional networks for text classification. In: *Advances in Neural Information Processing Systems* 28 (2015), arXiv:1509.01626